■ Wave Properties: Frequency, Wavelength & Amplitude

■ Grade 6 · Science	■ 45 minutes	■ Topic: Physics – Waves

■ Learning Objectives

By the end of this lesson, students will be able to:

- ■ Define what a wave is and explain how energy travels through waves.
- ■ Identify and describe the parts of a wave: crest, trough, and rest position.
- ■ Explain frequency and measure it in Hertz (Hz).
- ■ Define wavelength as the distance between two crests.
- ■ Describe amplitude and connect it to the energy a wave carries.
- ■ Compare wave properties using real-world examples.

■ What is a Wave?

A **wave** is a way energy moves from one place to another. Waves are all around us every day! Sound waves help us hear music and voices. Water waves create ripples when you toss a stone in a pond. Light waves let us see beautiful colors. Energy travels through these waves — but the material itself doesn't move far!

■ Sound Waves

Allow us to hear music, voices, and the world around us.

■ Light Waves

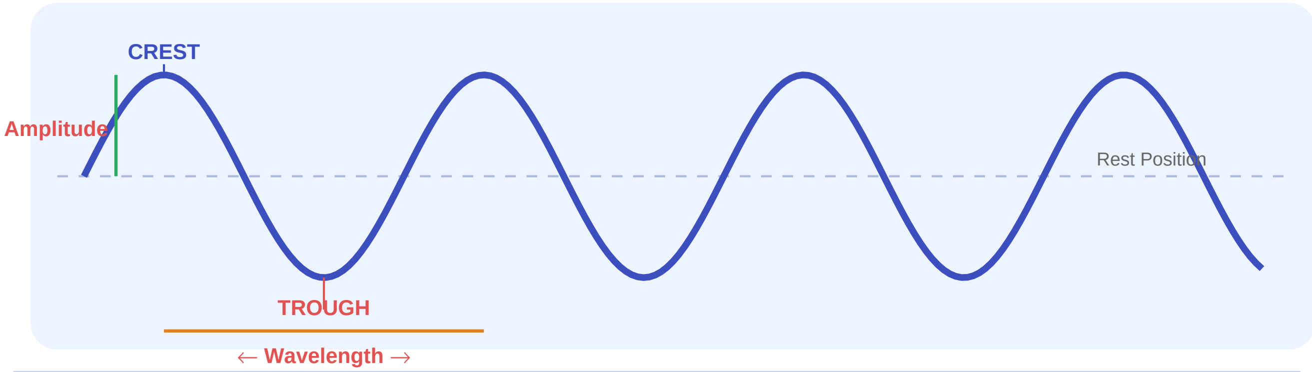
Let us see colors, rainbows, and the world around us.

■ Water Waves

Form ripples in ponds and powerful waves in the ocean.

■ Parts of a Wave

Every wave has special parts. When you look at a wave, you can see it goes up and down in a repeating pattern. Understanding these parts helps us describe how waves move energy.



Part	Description	Real-Life Example

Crest	The highest point of a wave — like the top of a hill.	The peak of an ocean wave.
Trough	The lowest point of a wave — like the bottom of a valley.	The dip between two ocean waves.
Rest Position	The middle line — where the wave would be if perfectly still.	Calm, flat water with no waves.

■ Frequency

What is Frequency?

Frequency is how many waves pass by in one second. Think of it like clapping your hands! Fast clapping = high frequency. Slow clapping = low frequency.

■ Unit: **Hertz (Hz)**

If 5 waves pass in 1 second → Frequency = **5 Hz**

High vs Low Frequency

■ High Frequency = High-Pitched Sound

Fast waves → whistle, bird chirping, flute.

■ Low Frequency = Low-Pitched Sound

Slow waves → drum, thunder, bass guitar.

■ Wavelength

Wavelength is the distance between two consecutive wave crests (or two troughs). Think of it as measuring from the top of one wave to the top of the very next wave.

■ Longer Wavelength

Waves are stretched far apart from each other.

Example: **Red light** has the longest wavelength in visible light.

■ Shorter Wavelength

Waves are squeezed close together.

Example: **Violet light** has the shortest wavelength in visible light.

■ Ocean vs Pond

Small ripples in a pond = short wavelength.

Huge ocean waves = long wavelength.

■ Amplitude

Amplitude is how tall a wave is — measured from the rest position (the middle line) to the crest. Think of it like how high you can jump! A bigger amplitude means a taller wave with more energy.

Amplitude	Energy Level	Sound Example	Water Example
Large / Big	■ More energy	Shouting loudly	Giant ocean wave
Small / Little	■ Less energy	Whispering softly	Tiny pond ripple

■ **Remember:** When you jump into a pool and make a big splash, the waves are larger (higher amplitude) because you transferred more energy into the water!

■ Quick Comparison: Wave Properties

Property	What it Measures	Unit	High means...	Low means...
Frequency	Waves per second	Hertz (Hz)	Many waves/sec → high pitch	Few waves/sec → low pitch
Wavelength	Distance between crests	Metres (m)	Waves far apart (e.g. red light)	Waves close together (e.g. violet)
Amplitude	Height of wave from rest	Metres (m)	More energy, louder sound	Less energy, softer sound

■ How Do We Measure Waves?

■ Measuring Frequency	■ Measuring Wavelength	■ Measuring Amplitude
Use a **stopwatch** and count how many waves pass a fixed point in exactly 1 second. More waves = higher frequency!	Use a **ruler** to measure from the top (crest) of one wave to the top of the very next wave.	Use a **ruler** to measure from the rest position (middle line) straight up to the crest.

■ Fun Experiments to Try!

Experiment	What You Need	What to Observe
■ **Water Tray Waves**	Tray, water, ruler	Gently tap one end. Measure the wavelength and amplitude with your ruler.
■ **Sound Frequency Game**	Whistle, drum (or any two sounds)	Listen carefully — can you guess which sound has higher frequency?
■ **Ripple Patterns**	Pebbles of different sizes, water	Drop pebbles of different sizes. Notice how bigger drops make larger-amplitude waves!

✏■ Check Your Understanding — Quick Quiz

1. What is a wave?

a) A particle that moves from one place to another
b) A way energy moves from one place to another ■
c) A solid object that vibrates
d) None of the above

2. A whistle has _______ frequency compared to a drum.

a) Lower
b) The same
c) Higher ■
d) No

3. What unit do we use to measure frequency?

a) Metres
b) Kilograms
c) Seconds
d) Hertz (Hz) ■

4. Wavelength is measured from:

a) Crest to trough
b) Trough to rest position
c) One crest to the next crest ■
d) Rest position to trough

5. A bigger amplitude means:

a) Less energy
b) More energy and a louder sound ■
c) Higher frequency
d) Shorter wavelength

■ Wave Properties Recap!

■ Frequency = How many waves pass per second (measured in Hz)
■ Wavelength = Distance between two wave crests
■ Amplitude = Height of wave from rest position to crest

"Waves carry energy — the bigger and faster, the more powerful!"